

Tahsin Reza

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EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka

Jan 2022 - June 2026

Bachelor of Urban and Regional Planning (BURP)

CGPA: 3.70/4.00

Relevant Coursework: Statistics I and II, Programming Techniques, GIS and Remote Sensing, Transportation Policy and Planning, Environmental Planning and Urban Planning Techniques

RESEARCH INTEREST

Traffic Modeling and Micro-simulation, Travel Behavior & Mode Choice, Data Science for Transport Planning, Geospatial Analytics for Transportation, Urban Mobility, Safety, Accessibility & Equity in Transportation

ACADEMIC HONORS AND AWARDS

- University Merit Scholarship, BUET (2023)

RESEARCH EXPERIENCE

[1] Time-Headway Analysis of Pedal Rickshaws and Battery-Operated Three-Wheelers in Dhaka's Non-Lane-Based Heterogeneous Traffic

- Built an automated extraction pipeline, a YOLO26 detector trained on roughly 207,000 self-labeled vehicle instances paired with BoT-SORT tracking, producing about 900 follower-leader observations across two corridors that use for both manuscripts below
 - (a) Pair-Specific Time Headway Distributions and Covariate-Conditioned AFT Models for Pedal Rickshaws and Battery-Operated Three-Wheelers in Dhaka's Non-Lane-Based Heterogeneous Traffic**
Manuscript Submitted (Transportation Research Board (TRB) Annual Meeting 2027, 1st Author) [\[Online Version\]](#)
 - Fitted pair-specific headway distributions and accelerated failure time (AFT) models to explain distributional characteristics and shifts, accounting for covariate effects that feed both simulation modeling and the policy horizon.
 - (b) Speed, Not Spacing: A Location-Scale Decomposition of Pedal- and Battery-Powered Three-Wheeler Following and Its Speed-Policy Implications in Dhaka**
Manuscript in Preparation
 - Built a maximum-likelihood location-scale regression of vehicle time headway with an exact kinematic-identity decomposition, applied to 898 video-extracted following events from two Dhaka corridors.
 - Battery operated three-wheelers followed 30–39% closer than pedal rickshaws; 66–80% of it a speed effect, not spacing; yielding 2.5× higher exposure to critically short following across both sites.

[2] Mono- versus Intermodal Trip Choice in Dhaka: Bayesian Multilevel Models and Explainable Machine Learning

Manuscript in Preparation

- Built a four-level (M0–M3) Bayesian multilevel logistic regression and an explainable Random Forest/SHAP pipeline to model monomodal vs. intermodal trip choice across 25,242 trips nested within 3,729 Dhaka households.
- Confirmed intermodality is necessity-driven, not discretionary: households below Tk 40,000/month income and in tinshed housing ($OR \approx 5.9$ vs. self-built housing) chained modes most, while most vehicle-ownership categories cut the odds by ~90% and higher income by ~75%, the reverse of the affluence-linked intermodality reported in European studies.

[3] Flow Characteristics of Traditional Rickshaws and Battery-Operated Three-Wheelers in Non-Lane-Based Heterogeneous Traffic of Dhaka

Bachelor's Thesis, Department of Urban and Regional Planning, BUET (2026) [\[Thesis Report\]](#)

- Conducted videographic surveys at three Dhaka corridors, manually extracting 531 valid target-leader time-headway observations using a reference-line method.
- Benchmarked six tree-based models and selected a tuned CatBoost model (Test $R^2 = 0.674$, MAE = 0.400 s) with SHAP analysis identifying speed as the strongest predictor, confirming battery-operated three-wheelers consistently follow with shorter headways than rickshaws - evidence against regulating them as one vehicle class.

SELECTED UNDERGRADUATE PROJECTS

[1] Analyzing Core Rural Road Networks and Land Use Conversion: A Case Study on Selected Upazilas in Bangladesh

Rural Planning Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed all unions across 14 upazilas via KIIs and GIS buffer analysis (LGED roads, BBS demographics, Sentinel land-use), then built a log-log regression model ($R^2=0.44$, $p<0.001$) of core road density. Population ($\beta=0.305$) and education-institution ($\beta=0.205$) density were strongest predictors. Core roads served ~90% of growth centers/markets, 50–60% of key facilities, amid 81% natural land-use conversion.

[2] A Study on Factors Influencing Access Mode Choice of Metro Rail (MRT) Users in Dhaka City

Social and Physical Survey Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 345 MRT users at 3 stations to identify first/last-mile access mode determinants. Rickshaw dominated (41%), walking/bus tied at 20%; 84%+ traveled under 5 km and about 77% of trips under 1 km were made on foot. Walking led in Pallabi (shortest distance), rickshaw in Uttara North (lower crime risk), and bus in Agargaon (consistent road width).

[3] Factors Influencing Pedestrian Catchment Area of Major MRT Stations in Dhaka

Transport Planning Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 717 commuters at Motijheel and Farmgate using random sampling and station-gate video counts, mapping built-environment variables across six GIS buffer rings (250–2000 m) and training a Random Forest Regressor to predict egress-walk distance. Mean distances were 440.8 m and 401.7 m, with 12–17% exceeding the conventional 800 m catchment radius; the model achieved $R^2 = 0.94$, with road density and floor area outweighing sociodemographic factors.

[4] Transportation Problem Solver: North-West Corner and Least Cost Methods [\[Presentation\]](#)

Programming Techniques, Department of Urban and Regional Planning, BUET

- Built a Python tool that computes and compares initial feasible solutions to the classical transportation problem using both the North-West Corner and Least Cost methods.

PROFESSIONAL EXPERIENCE

Assistant Urban Planner

July 2026 - Present

Sheltech Consultants (Pvt.) Ltd., Dhaka

- Drafted the transportation planning volume of the Pourashava master plan report under IUGIP (Improving Urban Governance & Infrastructure Program), based on the survey report and ToR.
- Refined the LOS determination methodology and supported the wider project team with transport-related queries while reporting for all Pourashavas under the program.

Intern

November 2025 - December 2025

Dhaka Transport Coordination Authority (DTCA), Dhaka

- Assessed the traffic impact of converting a road in Dhaka into a pedestrian precinct. Ran traffic and pedestrian surveys, then built and compared before-and-after microsimulation scenarios in PTV Vissim to test the proposed layout.

TEACHING EXPERIENCE

Tutoring Secondary and Higher Secondary students (Physics, Math, ICT, Chemistry)

2021-Present

- Provided academic support to over 12 students across varying levels
- Served as Mathematics instructor at a local coaching center

SOFTWARE AND PROGRAMMING SKILLS

Programming Language: Python, R, C | **GIS Tool:** ArcGIS, ArcGIS Pro, Google Earth Engine | **Transportation Modelling:** PTV Vissim | **Modeling and ML:** Discrete Choice Models, Survival/AFT Models, Tree-Based Ensemble Models, SHAP, YOLO Object Detection, scikit-learn, Distribution Fitting (scipy) | **Design and Drafting:** AutoCAD, Adobe Illustrator, Adobe Photoshop | **Productivity:** MS Word, Excel, PowerPoint, Project, CapCut

EXTRA-CURRICULAR ACTIVITIES

- Secretary, Student Chapter, **Bangladesh Institute of Planners (BIP)** (2025-2026)
- Vice President, **URP Students' Association of BUET (USAB)** (2025-2026)
- Joint General Secretary, **Help_BUETian** (2025-2026)

REFERENCES

Dr. Md. Musleh Uddin Hasan

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Bangladesh University of Engineering and Technology
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Relationship: Thesis Supervisor and Research Mentor

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Relationship: Course Instructor and Research Mentor

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